Xu-Cheng He

Academy Research Fellow / Group leader

University of Helsinki

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Education

University of Helsinki — *Finland*
PhD Atmospheric Science (Distinction) — *2018.01 - 2021.09*

University of Helsinki — *Finland*
MSc Atmospheric Science — *2015.08 - 2017.10*

Yunnan University — *China*
BSc Atmospheric Science — *2011.09 - 2015.07*

Professional Experience

2025.01-Present	**Academy Research Fellow and group leader (2025.09–); Academy post-doctoral researcher and sub-group leader (2025.01–2025.08)**, University of Helsinki
2017.09-Present	**CERN CLOUD collaboration: Principal Investigator (2026-); Co-Principal Investigator (2025-2026); Coordinator of marine experiments (2017-2024)**, CERN / University of Helsinki
2023.01-2024.12	**Visiting Researcher (simulation of aerosols)**, University of Cambridge
2022.03-2022.12	**Visiting Researcher (global model training)**, Carnegie Mellon University and Finnish Meteorological Institute

Awards

2025	**Airmodus best oral presentation award**, The 22nd International Conference on Nucleation and Atmospheric Aerosols	*€200*
2023	**Invited participant of the Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XVII)**, Brookhaven National Laboratory	
2022	**Extraordinary Potential Prize of 2021 Chinese Government Award for Outstanding Self-financed Students Abroad (20 recipients globally)**, Ministry of Education, China	*$10,000*
2022	**Doctoral dissertation prize, all faculties (4/500)**, University of Helsinki	*€4,000*
2022	**NOSA ECS Aerosologist award (Early Career Scientist)**, Nordic Society for Aerosol Research	
2021	**Outstanding thesis award**, Doctoral school in Natural Sciences, University of Helsinki	*€2,000*
2015	**International student grant**, University of Helsinki	*€1,500*
2015	**Best thesis award (1/43)**, Yunnan University	
2012	**Campus star in science and innovation (10/4000)**, Yunnan University	*¥2,000*

Fellowships & Grants

2027-2032	**European Research Council Starting Grant 2026, MARFORCE**, European Research Council	*€2,394,892*
2025-2028	**ERC application boost funding**, University of Helsinki	*€150,000*
2025-2029	**Academy research fellowship**, Research Council of Finland / University of Helsinki	*€1,126,320*
2025-2026	**co-PI, Constraining the size distribution and chlorine production of ferric chloride aerosols for quantitative atmospheric methane removal**, Spark Climate Solutions	*$299,248*
2022-2025	**Postdoctoral researcher fellowship**, Research Council of Finland / University of Helsinki	*€364,377*
2022	**Postdoctoral research grant**, Jenny and Antti Wihuri foundation	*€56,000*
2018-2021	**Doctoral school fellowship**, University of Helsinki	*ca. €108,000*

Key publications

Publications as a **CO-FIRST**[%] or **CORRESPONDING AUTHOR**[#]. Names in bold denote current or former members of my research group or collaborators whom I have directly mentored.

Summary: 6 manuscripts as (co-)first-author and 7 as last-author, in total 12 as corresponding author, including 2 in *Science* and 2 in *Nature*.

2026

A. Shcherbinin[#%], **J. Shen**[#%], O. Kausiala, ..., H. Finkenzeller, M. Rissanen, **X.-C. He**, Networked Open-Source Infrastructure for Scalable Analysis of Chemical Ionization Mass Spectrometry Data. **ChemRxiv preprint**. (2026).

W. Yu, **R. Baalbaki**, **J. Shen**, ..., I. Riipinen, **X.-C. He**[#], K. Lehtipalo, Impact of humidity on aerosol growth from methanesulfonic acid. **Environmental Science: Atmosphere**. (2026).

R. Baalbaki[%], **J. Shen**[%], M. Simon, ..., J. Kirkby, K. Lehtipalo[#], **X.-C. He**[#], Role of methanesulfonic acid in atmospheric particle nucleation and growth. **Nature**. (2026).

X.-C. He, N.L. Abraham[#], H. Ding, ..., J. Weber, A.T. Archibald[#], H. Gordon, Evaluation of UKESM aerosol size and composition using ATom measurements indicates missing marine aerosol formation mechanisms. **Atmospheric Chemistry and Physics**. (2026).

2025

R. Zhang, H.-B. Xie[#], F. Ma, R. Yin, J. Chen, **X.-C. He**, Quaternary Nucleation of Iodine and Sulfur Oxoacids in the Marine Atmosphere: Unexpected Role of Methanesulfonic Acid. **Journal of Geophysical Research: Atmosphere**. (2025).

2024

J. Shen[%], D.M. Russell[%], J. DeVivo,..., J. Kirkby[#], J. Curtius[#], **X.-C. He**[#], New particle formation from isoprene under upper tropospheric conditions. **Nature**. (2024).

B. Rörup, **X.-C. He**[#], J. Shen,..., R. Volkamer, D. Worsnop, K. Lehtipalo, Temperature, humidity, and ionisation effect of iodine oxoacid nucleation. **Environmental Science: Atmosphere**. (2024).

Y. Zhang[%], **D. Li**[%], **X.-C. He**[#],..., J. Jiang, A. Ding, M. Kulmala, Iodine oxoacids and their roles in sub-3 nanometer particle growth in polluted urban environments. **Atmospheric Chemistry & Physics**. (2024).

2023

X.-C. He[#], M. Simon, S. Iyer, H.-B. Xie[#], ..., N.M. Donahue, M. Sipilä[#], M. Kulmala[#], Iodine oxoacids enhance nucleation of sulfuric acid particles in the atmosphere. **Science**. (2023).

X.-C. He[#], **J. Shen**[#], S. Iyer,..., J. Mikkilä, M. Sipilä, J. Kangasluoma, Characterisation of gaseous iodine species detection using the multi-scheme chemical ionisation inlet 2 with bromide and nitrate chemical ionisation methods. **Atmospheric Measurement Techniques**. (2023).

F. Ma, H.-B. Xie[#], R. Zhang,..., M. Engsvang, J. Elm, **X.-C. He**[#], Enhancement of Atmospheric Nucleation Precursors on Iodic Acid Induced Nucleation: Predictive Model and Mechanism. **Environmental Science and & Technology**. (2023).

H. Finkenzeller[%#], S. Iyer[%], **X.-C. He**,..., T. Kurten[#], M. Rissanen, R.V. Volkamer[#], The gas-phase formation mechanism of iodic acid as an atmospheric aerosol source. **Nature Chemistry**. (2023).

2022

R. Zhang, H.-B. Xie[#], F. Ma,..., M. Sipilä, M. Kulmala, **X.-C. He**[#], Critical Role of Iodous Acid in Neutral Iodine Oxoacid Nucleation. **Environmental Science & Technology**. 56, 14166-14177 (2022).

2021

M. Wang[%], **X.-C. He**[%#], H. Finkenzeller, S. Iyer, D. Chen,..., M. Rissanen, R. Volkamer, Y. J. Tham[#], N. M. Donahue, M. Sipilä, Measurement of iodine species and sulfuric acid using bromide chemical ionization mass spectrometers. **Atmospheric Measurement Techniques**. 14, 4187-4202 (2021).

X.-C. He[#], Y. J. Tham, L. Dada, M. Wang, H. Finkenzeller,..., N. M. Donahue, R. Volkamer, J. Kirkby[#], D. R. Worsnop, M. Sipilä[#], Role of iodine oxoacids in atmospheric aerosol nucleation. **Science**. 371, 589–595 (2021).

X.-C. He[#], S. Iyer, M. Sipilä, A. Ylisirniö, M. Peltola,..., V.-M. Kerminen, R. C. Flagan, J. Kirkby[#], T. Kurtén, M. Kulmala, Determination of the collision rate coefficient between charged iodic acid clusters and iodic acid using the appearance time method. **Aerosol Science & Technology**. 55, 231–242 (2021).

Y. J. Tham, **X.-C. He**, Q. Li, C. A. Cuevas, J. Shen,..., M. Kulmala, C. O'Dowd, M. Dal Maso, A. Saiz-Lopez[#], M. Sipilä[#], Direct field evidence of autocatalytic iodine release from atmospheric aerosol. **Proceedings of the National Academy of Sciences**. 118 (2021).

2020

M. Wang[%], W. Kong[%], R. Marten, **X.-C. He**,..., J.H. Seinfeld, I. El-Haddad, R.C. Flagan, N.M. Donahue[#], Rapid growth of new atmospheric particles by nitric acid and ammonia condensation. **Nature**. 581 (2020).

2019

D. Zhao, R. Yang[#], Y. Tao, W.K. Zhang and **X.-C. He**, Objective detection of the Kunming quasi-stationary front. **Theoretical and Applied Climatology**. 138 (2019).

2017

S. Iyer[#], **X.-C. He**, N. Hyyttinen, T. Kurtén[#] and M.P. Rissanen, Computational and Experimental Investigation of the Detection of HO2 Radical and the Products of Its Reaction with Cyclohexene Ozonolysis Derived RO2 Radicals by an Iodide-Based Chemical Ionization Mass Spectrometer. **The Journal of Physical Chemistry A**. 121 (2017).

F. Bianchi[#], O. Garmash, **X.-C. He**,..., M. Kulmala, M. Ehn and H. Junninen, The role of highly oxygenated molecules (HOMs) in determining the composition of ambient ions in the boreal forest. **Atmospheric Chemistry and Physics**. 17 (2017).

Total publication count: 72. Total citations: Researchgate (3,394), Google scholar (3,546).

Presentations

Selected Invited Talks

2025.12 - 2026.01. Invited seminars, **Dalian University of Technology, Nanjing University, Chinese Academy of Sciences**, China

2025.05. Invited seminar, **University of Reading**, UK

2025.03. Invited seminar, **University of Cambridge**, UK

2025.02. Invited seminar, **California Institute of Technology**, USA

2024.06. Invited seminar, **University of Helsinki**, Finland

2024.05. Invited seminar, **Max Planck Institute for Chemistry**, Germany

2024.04. Invited seminar, **Massachussetts Institute of Technology**, USA

2024.04. Invited seminar, **SOLAS open seminar series**

2024.03. Invited seminar, **University of Cambridge**, UK

2024.01. Invited seminar, **University of Birmingham**, UK

2022.08. Invited speaker, **Gordon Research Conference**, Italy

2022.05. Invited talk, **Carnegie Mellon University**, USA

2021.01. Invited talk, **Nanjing University**, China

Contributed Presentations

American Geophysical Union annual meeting (session convener, 2026)

American Meteorological Society annual meeting (session co-chair, 2026, 2025, 2024, 2023)

Atmospheric Chemistry Colloquium for Emerging Senior Scientists (Invited, 2023)

Gordon Research Conference - Molecular and Ionic Clusters (Invited talk, 2022)

Gordon Research Conference - Atmospheric Chemistry (Poster, 2025, 2023)

European Geosciences Union General Assembly (Talk, 2021, 2020)

International Aerosol Conference (Talk, 2026, 2023; Poster, 2017)

European Aerosol Conference (Talk, 2024, 2022; Poster, 2019)

International Conference on Nucleation and Atmospheric Aerosols (Talk, 2025, 2023; Poster, 2017)

International Conference on Aerosol Cycle (Talk, 2017)

Free Radical Symposium (Poster, 2017)

Surface Ocean Lower Atmosphere Study (Poster, 2019)

Cryosphere and Atmospheric Chemistry (Poster, 2017)

Teaching

2026	**Summer School on Formation and Growth of Atmospheric Aerosols**, Lecturer	*University of Helsinki*
2024	**Tropospheric Atmospheric Chemistry**, Guest Lecturer	*University of Helsinki*
2024	**Synthesis of physical chemistry, experiments, observations and models to understand atmospheric particle formation and climate impact**, Lecturer	*University of Helsinki*
2019	**Climate science at high latitudes: eScience for linking Arctic measurements and modeling**, Teaching Assistant	*University of Helsinki*
2018	**Formation and growth of atmospheric aerosols**, Teaching Assistant	*University of Helsinki*

Supervision & Mentoring

Current group member: Jiali Shen (University Researcher), Dejia Yin (Post-doc), Julia David (Post-doc), Wenjuan Yu (PhD researcher)

PhD supervision/mentoring: Jiali Shen (2023.04 with a distinction), Rima Baalbaki (2024.06), Birte Rörup (2024.06), Wenjuan Yu (on-going)

PhD thesis committee: Yifan E (on-going)

MSc supervision/mentoring: Ying Zhang (2022.07), Duzitian Li (2024.07)

BSc supervision/mentoring: Duzitian Li (2021.07)

Service, outreach & Professional Development

Conference chair

2022-2026: Formation and impacts of atmospheric aerosols and cloud condensation nuclei: experiment, observation, and modeling. Co-chair. American Meteorological Society Annual Meeting.

2025: Young Principal Investigators session. Chair. International Conference on Nucleation and Atmospheric Aerosols.

2026: Aerosols, clouds and their precursors in the marine atmosphere. Co-chair. International Aerosol Conference 2026.

2026 - : From Molecules to Climate: Process-Level Understanding of Aerosol and CCN Formation. Convener. American Geophysical Union Annual Meeting.

2027 - : Aerosols and their impacts in the marine environment. Convener. European Geosciences Union General Assembly.

Manuscript Peer Review

Science, Nature Geoscience, One Earth, Environmental Science & Technology, EST Letters, EST Air, Geophysical Research Letters, Journal of Geophysical Research: Atmospheres, Atmospheric Chemistry and Physics.

Funding Review

National Science Foundation (USA)

Service

2023-present: Coordinator for the modelling group at the CLOUD project, CERN

2017-present: Run leader for the marine and polar aerosol formation program at the CLOUD project, CERN

2011-2012: Staff Member, Student Union Office (2011); Deputy Director, Student Union Office (2012)

Public engagement

2025.10: Delivered a tutorial lecture and guided tour of the CLOUD experiment to a visiting Finnish university delegation.

Media Coverage

Dec 2023	**Chemistry World**, Iodine compounds accelerate cloud formation over oceans and the poles
Oct 2021	**Sciencepost**, L'émission d'iode par l'océan, une influence inattendue sur la banquise arctique
June 2021	**Lab Manager**, The Impact of Clouds on Climate Change
Feb 2021	**The Atlantic**, The Arctic Has a Cloud Problem
Feb 2021	**SCIENMAG**, Climate research: rapid formation of iodic particles over the Arctic
Feb 2021	**PHYS.ORG**, CLOUD at CERN reveals the role of iodine acids in atmospheric aerosol formation
Feb 2021	**ScienceDaily**, How iodine-containing molecules contribute to the formation of atmospheric aerosols, affect climate